

NovaPay REST API Gateway v3.2 - Technical Specification

Section 1: Authentication and Rate Limiting

All API requests must be authenticated using an mTLS certificate combined with an HMAC-SHA256 signature in the X-Nova-Signature header. The default rate limit for production credentials is 2 500 requests per minute per IP CIDR block. Exceeding this quota triggers an immediate HTTP 429 response.

Table 1: System Error Codes and Troubleshooting Protocol

Error Code	HTTP Status	Error Identifier	Action Required / Description
ERR_0x8001	401 Unauthorized	INVALID_KEY_ID	API key not found or expired in KMS
ERR_0x8004	422 Unprocessable	INSUFFICIENT_FUNDS_RESERVE	Merchant escrow account balance below required threshold
ERR_0x8019	409 Conflict	IDEMPOTENCY_KEY_COLLISION	Same Idempotency-Key used with different payload checked
ERR_0x9102	503 Unavailable	CLEARING_CIRCUIT_BREAKER	External banking clearing house timeout (> 3500ms)
ERR_0x9999	500 Internal	ENCRYPTION_HSM_FAILURE	Hardware Security Module synchronization fault

Retry Policy: For error ERR_0x9102 client libraries MUST implement exponential backoff with full jitter: $t = \min(t_{max}, t_{base} * 2^{attempt}) \pm \text{random}(0, 500ms)$. Do NOT retry ERR_0x8004 automatically.